

**MILESTONE 2 · DUE 20 SEPTEMBER 2026 · 40%****Final Implementation Plan and Working Prototype****TECHNICAL DESIGN, IMPLEMENTED MODULES AND RESULTS**

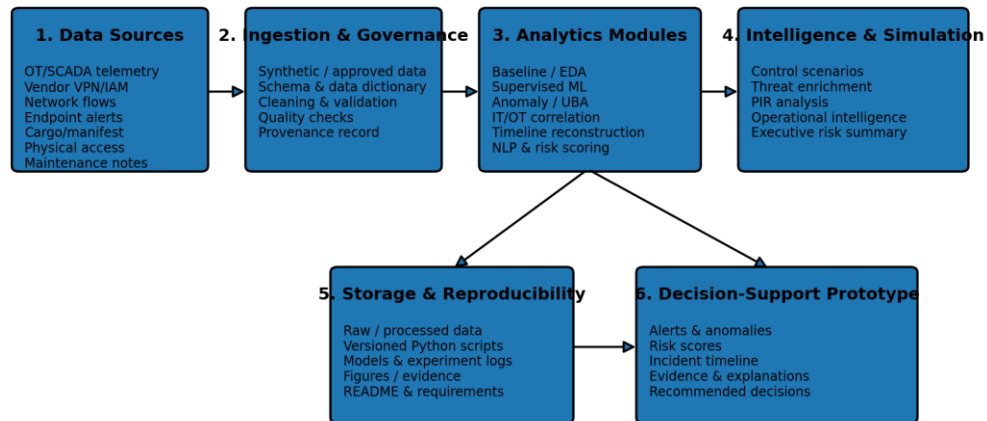
<https://github.com/mwandingim/Final-Implementation-Plan-and-Working-Prototype-215103238-AND-216086728.git>

1. Project status summary

Field	Response
Topic and project title	T21 · Maritime Port and Cargo-Terminal Cyber-Physical Security Analytics
Member A / Member B	215103238 Mwandingi — Data and Modelling Lead / 216086728 Kompeli — Security Engineering and Intelligence Lead
Overall completion estimates	About 90% complete. Every analytical module, the working prototype and internal QA are finished.
Working prototype location	T21_Working/07_dashboard_or_prototype/t21_analyst_prototype.html
Known blockers	There is no outstanding technical blocker.

2. Updated architecture and data flow

The architecture we built follows the design set out in our project charter. Seven synthetic sources OT/SCADA telemetry from the crane and yard systems, vendor VPN/IAM logs, network flow data, endpoint alerts, cargo/manifest events, physical-access records and maintenance notes feed into a governed ingestion and validation layer. From there the data flows through baseline analysis, the supervised and anomaly/access models, incident correlation, intelligence generation, simulation, NLP and predictive scoring. All outputs are kept in the project folder structure and surfaced through the analyst-facing prototype, so an analyst at the terminal can trace a finding all the way back to the raw evidence behind it.



Decision supported: determine whether anomalous vendor remote activity and yard/crane events require escalation, containment and continuity controls.

Figure 1. Updated T21 security analytics architecture and data flow.

Architecture implementation status

Architecture component	Status	Evidence / next action
Data sources	IMPLEMENTED	Synthetic T21 datasets, source inventory and provenance are complete.
Ingestion & governance	IMPLEMENTED	Validation, schema, duplicate and quality checks are complete.
Analytics modules	IMPLEMENTED	Baseline, supervised, anomaly, incident, intelligence, NLP and predictive modules are complete.
Storage & reproducibility	IMPLEMENTED	Scripts, datasets, models, outputs and documentation are retained in the project structure.
Simulation & intelligence	IMPLEMENTED	Control simulation, intelligence products and sensitivity analysis are complete.
Decision-support prototype	IMPLEMENTED	The integrated HTML analyst prototype is generated from the analytical outputs.

This mirrors what we committed to in Section 8 of our charter, so the implementation can be checked directly against what we originally proposed.



IMPLEMENTED DATA FLOW

3. Data inventory and quality

Dataset/source	Rows/volume	Date range	Quality issues	Cleaning/transformation	Status
Vendor VPN/IAM	1,800	2026-06-01 to 2026-06-30 UTC	No duplicate rows/event IDs; incident_id blank for normal events by design	Timestamp validation; duplicate checks; no destructive transformation	READY FOR EDA
OT/SCADA/crane telemetry	2,016	2026-06-01 to 2026-06-30 UTC	No duplicate rows/event IDs; incident_id blank for normal events by design	Timestamp validation; duplicate checks; command/label fields retained	READY FOR EDA
Network flows	2,200	2026-06-01 to 2026-06-30 UTC	No duplicate rows/event IDs; incident_id blank for normal events by design	Timestamp validation; duplicate checks; byte/flow fields retained	READY FOR EDA
Endpoint alerts	1,200	2026-06-01 to 2026-06-30 UTC	No duplicate rows/event IDs; incident_id blank for normal events by design	Timestamp validation; duplicate checks; alert/process fields retained	READY FOR EDA
Cargo/manifest events	1,204	2026-06-01 to 2026-06-30 UTC	No duplicate rows/event IDs; incident_id blank for normal events by design	Timestamp validation; duplicate checks; cargo/manifest fields retained	READY FOR EDA
Physical access	1,004	2026-06-01 to 2026-06-30 UTC	No duplicate rows/event IDs; incident_id blank for normal events by design	Timestamp validation; duplicate checks; badge/gate/zone fields retained	READY FOR EDA
Maintenance notes / text	304	2026-06-01 to 2026-06-30 UTC	No duplicate note IDs; incident_id	Timestamp validation; text retained for later NLP; no personal data by design	READY FOR NLP



Dataset/source	Rows/volume	Date range	Quality issues	Cleaning/transformation	Status
			blank for normal notes by design		
Asset inventory	28 assets	Static context	No missing values detected	Schema and uniqueness checks; context table retained	READY FOR JOINING
Integrated security index	9,728 derived event records	2026-06-01 to 2026-06-30 UTC	incident_id blank for normal events by design	Derived cross-source index; binary label mapping validated	READY FOR MODELLING
Scenario catalogue	5 scenarios	Static reference	No missing values detected	Schema check only	READY FOR SIMULATION

4. Baseline and exploratory findings

We completed the baseline and exploration work across all seven primary sources before training either model. The point of this stage was simply to understand what 'normal' looks like for vendor access and event volume at the terminal, so that later deviations actually mean something.

4.1 Security evidence baseline

After validation we were left with 9,728 primary event records spread across the seven sources. We also built an integrated security-event index by combining these sources, but we don't count that as an eighth source since it's derived, not collected. The 304 maintenance notes are kept separate and used only in the NLP stage.

Evidence source	Records	Non-normal events	Non-normal rate
Vendor VPN/IAM	1,800	106	5.89%
OT/SCADA/crane	2,016	113	5.61%
Network flows	2,200	160	7.27%
Endpoint alerts	1,200	96	8.00%
Cargo/manifest	1,204	44	3.65%
Physical access	1,004	58	5.78%
Maintenance notes	304	68	22.37%

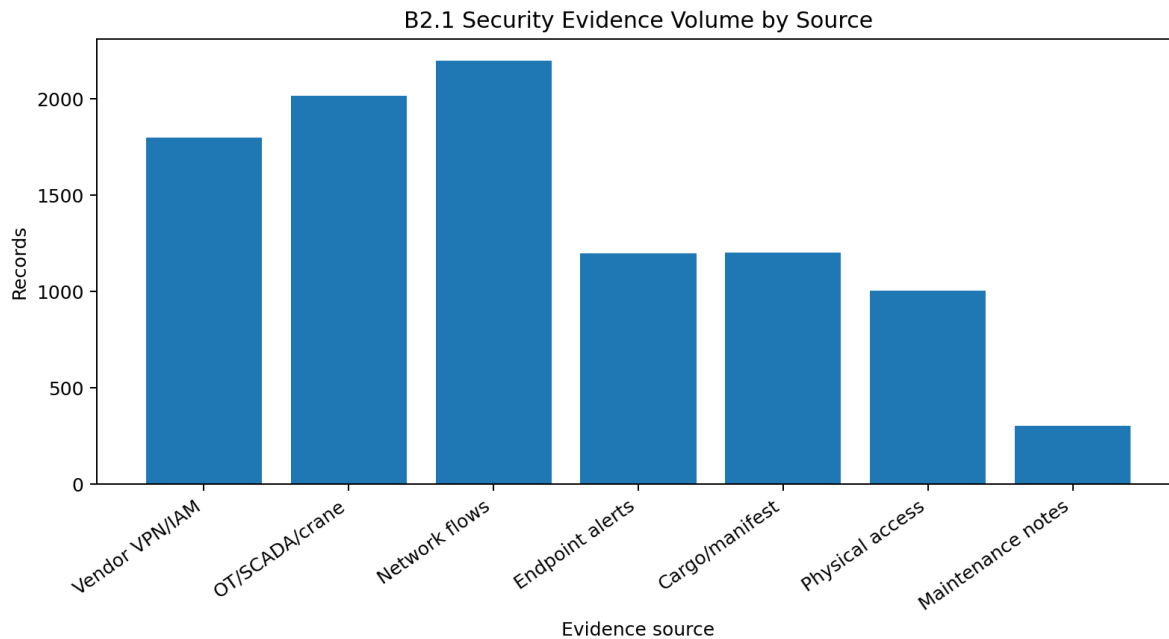


Figure 2. Security evidence volume by source.

4.2 Individual vendor baseline

We used vendor VPN/IAM activity to build an individual baseline for each vendor: session volume, non-normal event rate, off-hours activity, new-device activity, failed logins and median session length. V014 stood out with the highest non-normal rate at 13.19% (12 of 91 sessions), with V019 close behind at 12.05% (10 of 83) and V006 at 11.34% (11 of 97). On their own, these numbers only flag who deserves a closer look they don't prove anything about compromise.

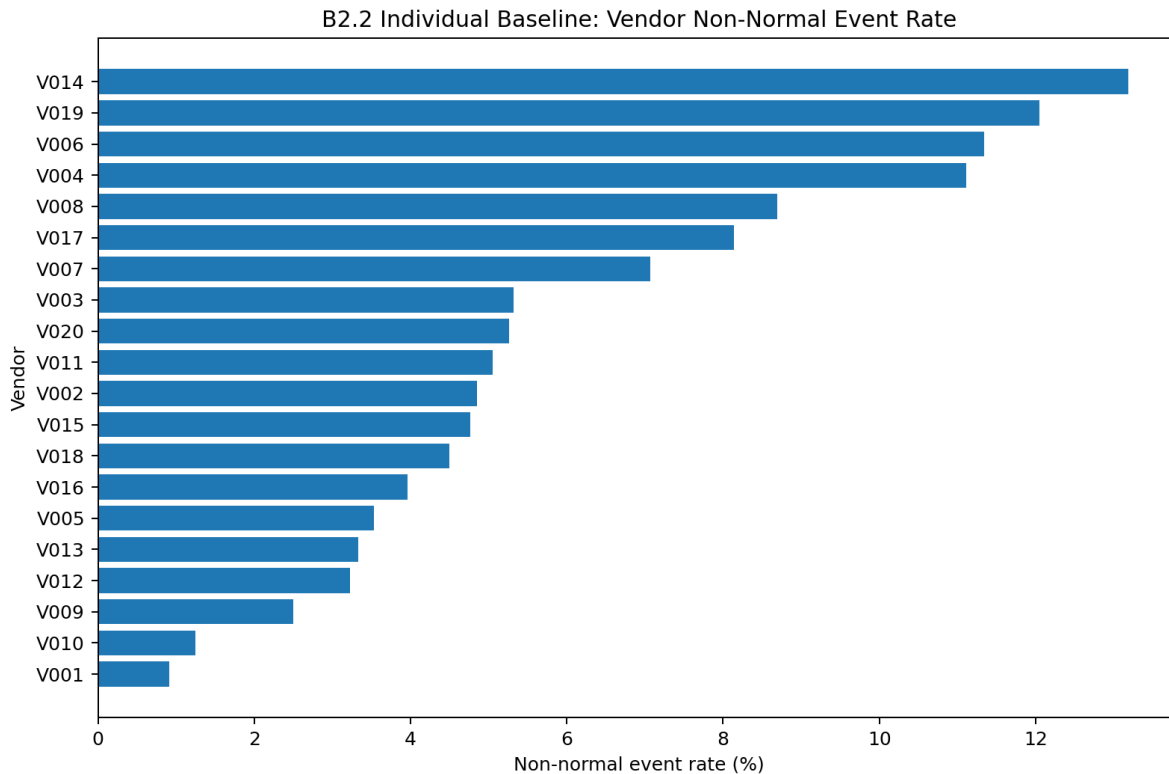


Figure 3. Individual vendor baseline based on non-normal session rate.

4.3 Peer-group baseline

We also grouped vendors by operational role to build a peer baseline, since comparing a crane vendor's behaviour against a network vendor's baseline wouldn't be fair. NetworkVendor came out highest at 6.99% (27 of 386 sessions), just ahead of MaintenanceVendor at 6.96% (22 of 316), while CraneVendor sat lowest at 4.22% (17 of 403). This role-based comparison lets us judge whether a vendor's deviation is genuinely unusual, rather than just typical for that kind of access.

Vendor role	Sessions	Non-normal events	Non-normal rate
NetworkVendor	386	27	6.99%
MaintenanceVendor	316	22	6.96%
AutomationVendor	353	22	6.23%
YardVendor	342	18	5.26%
CraneVendor	403	17	4.22%

4.4 Temporal access baseline

Looking at the hourly VPN pattern, suspicious sessions cluster more heavily in off-hours and late evening than normal ones do, so we kept off-hours timing as a candidate feature going into modelling. We're careful



not to over-read this: maintenance crews at a working port terminal genuinely do work outside normal daytime hours, so timing alone is contextual evidence, not proof of anything malicious.

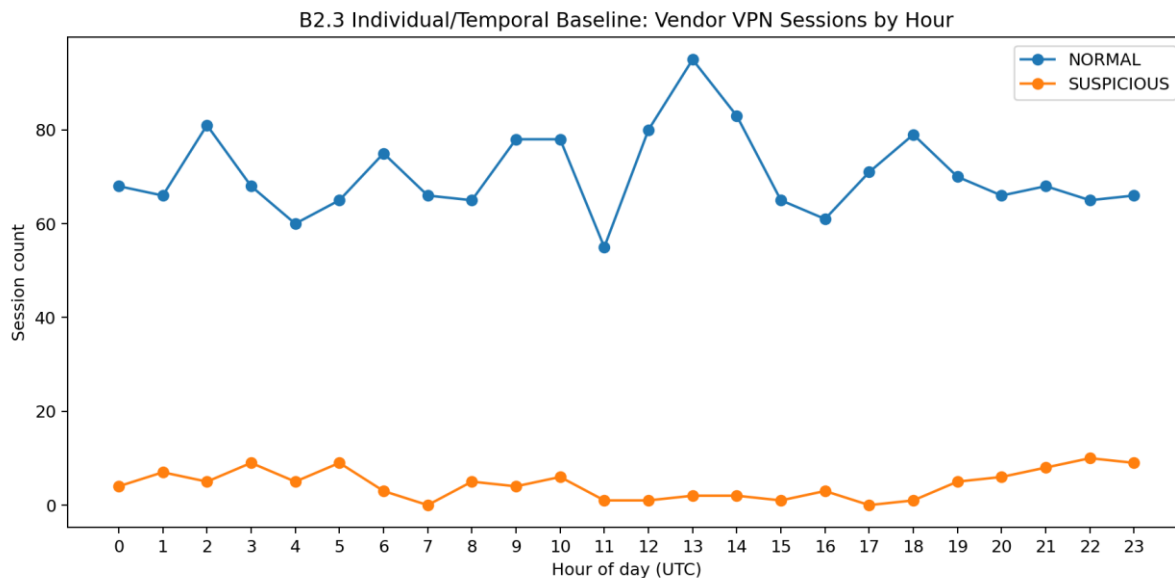


Figure 4. Vendor VPN session distribution by hour of day (UTC).

4.5 Baseline findings and analytical implications

Baseline dimension	Observed group/value	Finding	Analytical use
Individual baseline	V014	13.19% non-normal session rate; 12 of 91 sessions	Use vendor-level deviation as an investigation feature.
Peer-group baseline	NetworkVendor	6.99% non-normal rate; 27 of 386 sessions	Compare vendors against role-level behaviour before escalation.
Temporal baseline	Off-hours / late evening	Suspicious sessions are more concentrated outside normal hours	Use off-hours as a contextual risk indicator.
Device novelty	New device	New-device events occur in the suspicious subset of the synthetic dataset	Retain new-device status as a candidate model feature.

Together, these baseline distributions became the reference point for both the supervised model and the anomaly/access model that follow the behavioural features validated here are exactly what feeds into Section 5.



5. Supervised security model

5.1 Model design and target

Our supervised model classifies each vendor VPN/IAM session as either NORMAL (0) or NON-NORMAL (1), where NON-NORMAL folds together the SUSPICIOUS and INCIDENT labels from the dataset. Working from the full 1,800-session VPN/IAM dataset gave us enough behavioural signal to build an account- and vendor-level risk classifier.

Feature-wise, the model looks at vendor and account identity, authentication result, MFA status, session duration, role, access zone, off-hours status, new-device status, and derived time features (hour and day of week). We deliberately excluded direct identifiers and anything that reveals the outcome event label, incident, event id so the model can't cheat by learning the label instead of the behaviour.

5.2 Training and validation design

We split the data 70/30 using a fixed seed (210821) so the split is reproducible: 1,260 sessions for training, 540 held out for testing. Categorical fields were one-hot encoded and numeric fields kept as-is. The classifier is a Random Forest 300 trees, max depth 10, minimum leaf size 2, with balanced class weights to account for how rare the non-normal sessions are.

We used the standard 0.50 threshold for classification. The same held-out 30% was also reused as an unseen scoring set, and we exported the top 15 highest-risk sessions from it to feed directly into the incident-investigation stage.

5.3 Model evaluation results

On the held-out test set the model produced 507 true negatives, 1 false positive, 7 false negatives and 25 true positives 98.52% accuracy, 96.15% precision, 78.13% recall, and an F1-score of 86.21%, with a false-negative rate of 21.88%.

Metric	Result
Accuracy	98.52%
Precision	96.15%
Recall	78.13%
F1-score	86.21%
False-negative rate	21.88%
False positives / negatives	1 / 7
True positives / negatives	25 / 507



Figure 5. Confusion matrix for the held-out supervised-model test set.

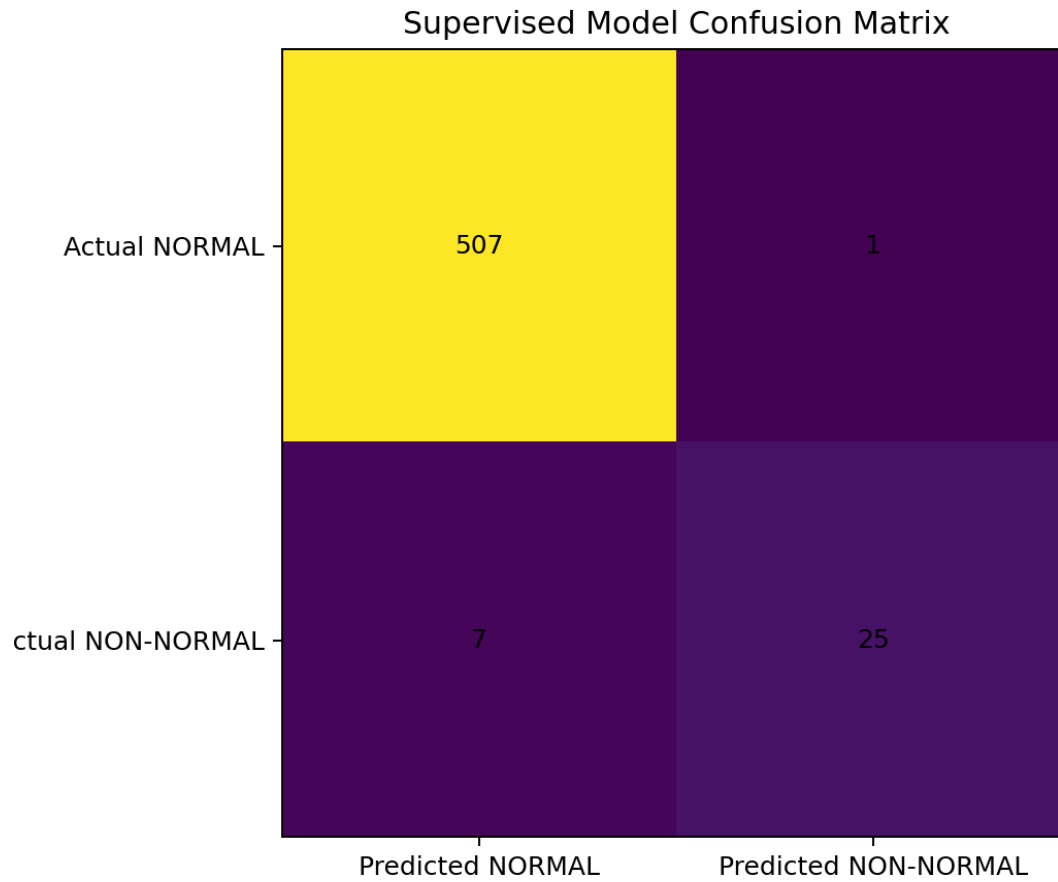
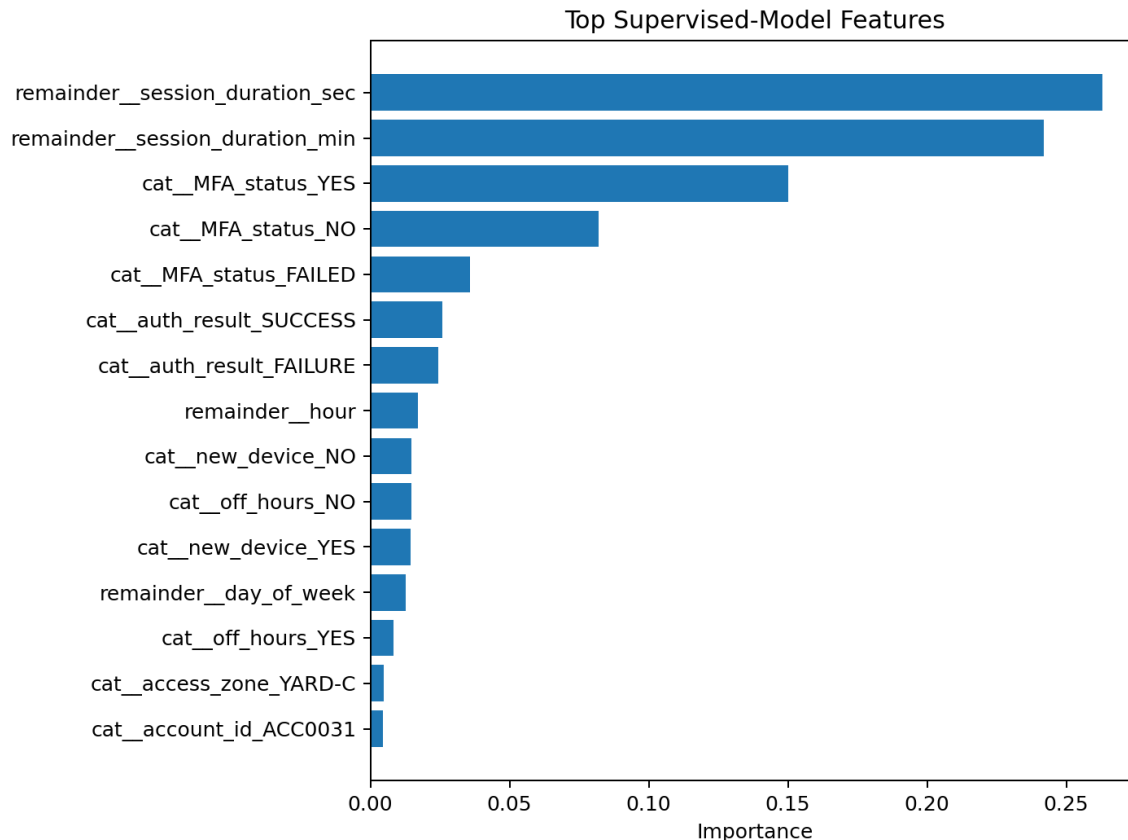




Figure 6. Top feature importance from the Random Forest model.



5.4 Threshold and security interpretation

Accuracy alone doesn't tell the real story here. For detecting a compromised account, recall and the false-negative rate matter more, because a false negative means a genuinely non-normal session slips through unflagged. At the 0.50 threshold the model caught 25 of 32 non-normal sessions (78.13% recall) and missed 7. Precision was strong at 96.15% only one normal session was wrongly flagged in the whole test set. We want to be upfront that these numbers describe performance on this synthetic dataset, not a guarantee of how the model would behave on real port traffic.

5.5 Investigation scoring and findings

The held-out sessions were ranked by predicted non-normal probability. The exported top-15 list contains the highest-risk vendor sessions for analyst follow-up. All 15 top-ranked sessions in this test partition had an actual non-normal label, providing a focused starting point for cross-source investigation. The ranking identifies repeated high-risk activity involving vendors V014, V017, V019, V011 and V004, among others. These observations are analytical leads and require correlation with OT, network, endpoint, cargo and physical-access evidence before an incident conclusion is made.

Output: 08_outputs/supervised_model/top_15_high_risk_sessions.csv



5.6 Reproducibility

The complete implementation is recorded in 03_notebooks_or_scripts/05_supervised_model.py. Model metadata, evaluation metrics, confusion matrix, feature-importance output, held-out scoring results and the top-15 investigation ranking are stored under 08_outputs/supervised_model/ and 04_models/supervised/.

6. Unsupervised, anomaly or access-analytics method

For the anomaly side, we trained an Isolation Forest on the same 1,800 vendor VPN/IAM sessions, but this time without giving it the event labels those were kept back purely for checking the results afterwards. Isolation Forest works by seeing how easily an observation can be separated from the rest through random partitioning; the easier it is to isolate, the more abnormal it's considered.

6.1 Entity, baseline and feature design

The unit of analysis is still the individual session. Features include session duration, authentication failure, missing MFA, off-hours access, new-device activity, cyclical hour-of-day encoding, role and access zone. As with the supervised model, we stripped out event/incident identifiers and labels before fitting, to avoid leaking the answer into the features.

Feature group	Implemented variables	Purpose
Session behaviour	session duration	Identify unusually short or long sessions.
Authentication	authentication failure, MFA missing	Capture abnormal authentication conditions.
Device/context	new device	Identify access from a device not previously associated with the session context.
Time	off-hours, hour sine/cosine	Represent unusual access timing without a discontinuity between 23:00 and 00:00.
Access context	role, access zone	Capture unusual combinations of vendor role and operational zone.

6.2 Anomaly model and parameters

We trained the Isolation Forest with 300 trees and seed 210821, fixing contamination at 0.05 before looking at the labels that setting alone produces a 5.00% flag rate. For readability we flipped the sign of scikit-learn's decision function so higher values mean more abnormal, since scikit-learn's default is the opposite (lower = more abnormal).

Parameter	Value
Algorithm	Isolation Forest
Sessions analysed	1,800
Number of trees	300
Contamination	0.05



Random seed	210821
Model fitting labels	Not used
Anomaly threshold	Isolation Forest contamination threshold

6.3 Anomaly detection results

90 of the 1,800 sessions (5.00%) came back flagged as anomalous. Checking that against the synthetic ground-truth labels afterwards gave precision of 0.9222, recall of 0.7830 and F1 of 0.8469 evaluation numbers only, since the model itself never saw the labels during training.

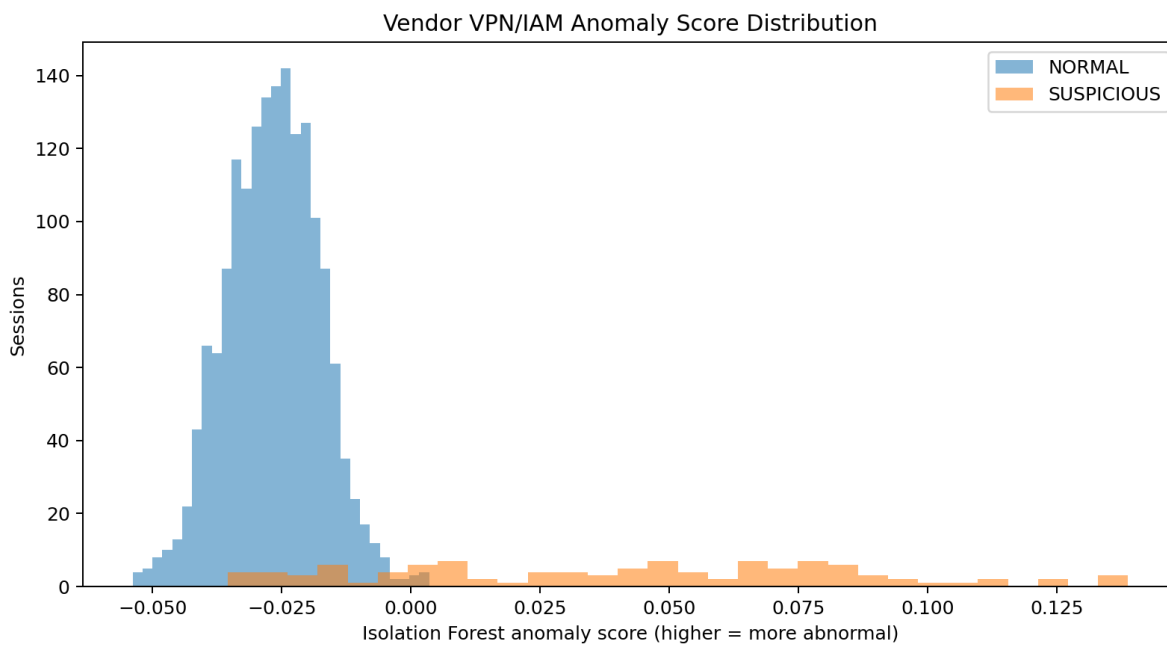
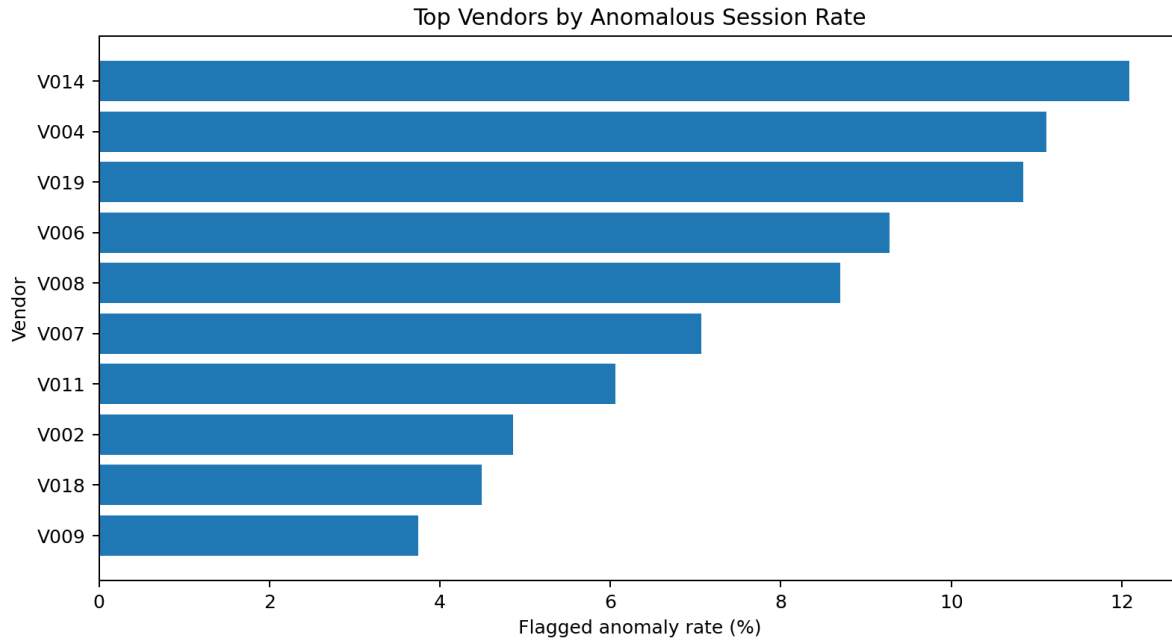
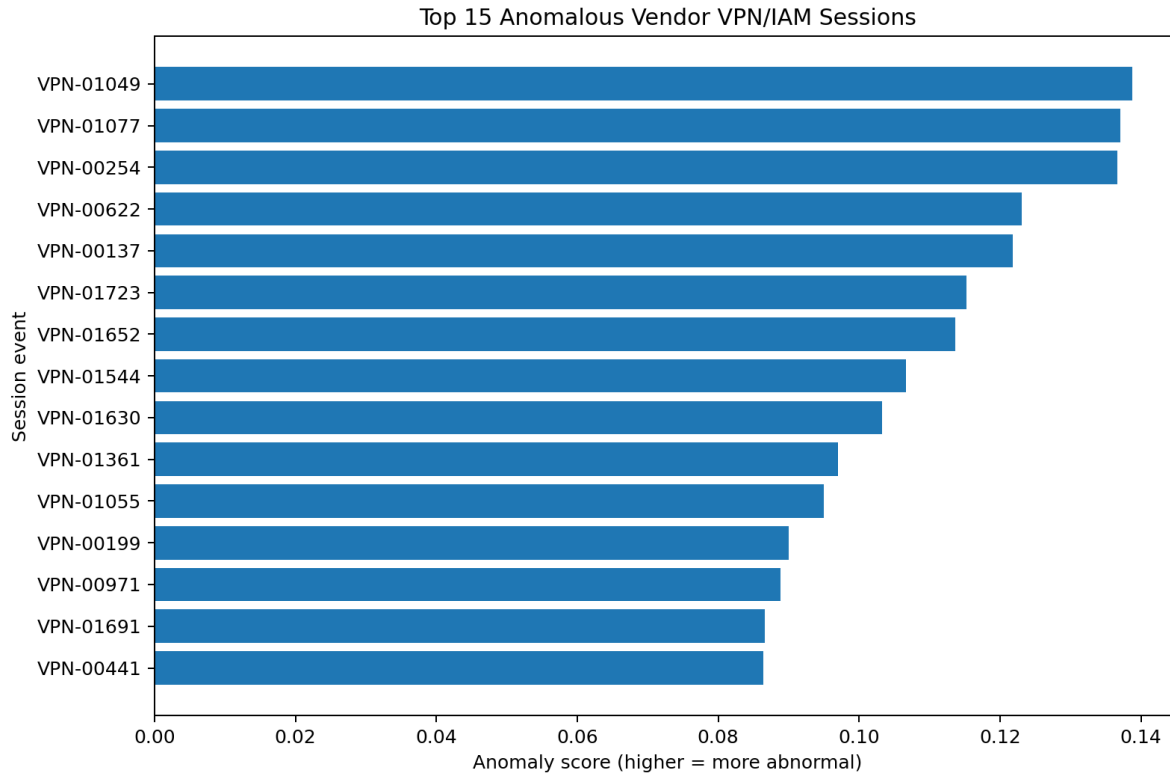




Figure 7. Isolation Forest anomaly-score distribution for vendor VPN/IAM sessions.



**Figure 8. Vendors with the highest anomalous-session rates.****Figure 9. Top 15 anomalous vendor VPN/IAM sessions.**

6.4 Highest-risk anomaly findings

Event	Vendor	Account	Anomaly score	Label	Off-hours	New device
VPN-01049	V019	ACC0058	0.1387	SUSPICIOUS	YES	YES
VPN-01077	V011	ACC0044	0.1371	SUSPICIOUS	NO	YES
VPN-00254	V019	ACC0058	0.1367	SUSPICIOUS	YES	YES
VPN-00622	V011	ACC0044	0.1231	SUSPICIOUS	NO	YES
VPN-00137	V019	ACC0058	0.1218	SUSPICIOUS	YES	YES
VPN-01723	V014	ACC0014	0.1153	SUSPICIOUS	YES	YES
VPN-01652	V008	ACC0050	0.1136	SUSPICIOUS	NO	YES
VPN-01544	V011	ACC0044	0.1066	SUSPICIOUS	NO	YES
VPN-01630	V006	ACC0031	0.1032	SUSPICIOUS	YES	YES
VPN-01361	V019	ACC0058	0.0970	SUSPICIOUS	YES	YES

The sessions ranked highest by the model share a pattern: new device access combined with off-hours activity. Encouragingly, the first ten sessions in the ranking were all labelled SUSPICIOUS in the underlying dataset, which tells us the unsupervised method is genuinely picking up the behaviour we



injected not just noise. That said, an anomaly flag is still a signal, not a verdict; legitimate emergency or maintenance access can look unusual too.

6.5 Comparison with the supervised model

The anomaly ranking was compared with the previously developed Random Forest investigation scoring. Three sessions occurred in both top-15 lists: VPN-00137, VPN-01077 and VPN-01691. The rank correlation between the anomaly ranking and the supervised model risk probability across the shared investigation records was -0.2735 when anomaly rank was used directly; this reflects the opposite direction of the two ranking variables and should not be interpreted as a strong agreement measure. The overlap itself shows that the two approaches identify some common sessions while also surfacing different evidence.

Method	Primary purpose	Strength	Limitation
Random Forest	Classified known non-normal behaviour	Uses labelled patterns and provides probability-based ranking.	Depends on representative labelled training data.
Isolation Forest	Finds unusual access behaviour without labels	Can surface previously unlabelled or unexpected patterns.	Flagged as anomalous is not the same as flagged as malicious.

6.6 Operational interpretation

In practice, we treat the anomaly output as one more triage signal, not a decision on its own. A high anomaly score is only worth acting on once it's checked against authentication evidence, endpoint/network activity, OT events and maintenance records. The real value shows up when the anomaly result is cross-referenced with the incident timeline, rather than read in isolation.

6.7 Reproducibility

The implementation is recorded in 03_notebooks_or_scripts/07_anomaly_access_analytics.py. The fitted model is stored as 04_models/anomaly/isolation_forest_vendor_access.joblib. Outputs are stored in 08_outputs/anomaly_access/, including the anomaly metrics, vendor summary, top-15 anomaly list, comparison with supervised scoring and three figures.

7. Incident investigation and response

The incident-investigation module pulls together correlated activity across identity/access, endpoint, OT/SCADA, network, physical-access, cargo and maintenance evidence all tied together through the synthetic dataset's incident identifiers. We investigated one case in full depth, INC-T21-001, and summarised the other three injected scenarios for coverage.

Incident	Scenario	Start (Utc)	End (Utc)	Events	Evidence Sources
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INC-T21-001	S01 – Compromised vendor account	2026-06-05T22:38:00+00:00	2026-06-05T22:59:00+00:00	37	Cargo, Endpoint, Maintenance notes, Network, OT/SCADA, Physical access, VPN/IAM
INC-T21-002	S02 – Unauthorized OT command sequence	2026-06-12T01:15:00+00:00	2026-06-12T01:34:00+00:00	37	Cargo, Endpoint, Maintenance notes, Network, OT/SCADA, Physical access, VPN/IAM
INC-T21-003	S03 – Remote administration and data transfer	2026-06-18T23:05:00+00:00	2026-06-18T23:22:00+00:00	29	Cargo, Endpoint, Maintenance notes, Network, OT/SCADA, Physical access, VPN/IAM
INC-T21-004	S04 – Cargo/manifest manipulation	2026-06-24T19:31:00+00:00	2026-06-24T19:49:00+00:00	34	Cargo, Endpoint, Maintenance notes, Network, OT/SCADA, Physical access, VPN/IAM

7.1 Incident evidence and scope

INC-T21-001 is our detailed case study. It runs from 22:38 to 22:59 UTC on 5 June 2026 and pulls in 37 correlated records spanning all seven evidence sources. In the scenario catalogue this maps to S01, “Compromised vendor account.” We treat that label as the dataset's ground truth for checking our work, but the analysis below is built purely from what the cross-source evidence actually shows.



The identity and device at the centre of this incident are V014 / ACC0014 / DEV0099. The evidence trail also pulls in CRANE03, vendor badge BADGE-0014 and cargo record CONT-004821.

7.2 Correlated incident timeline

The timeline below is the primary investigation output. Each row records the event identifier, timestamp, evidence source, observed action and analytical interpretation. This structure supports chronological reconstruction and cross-source correlation.

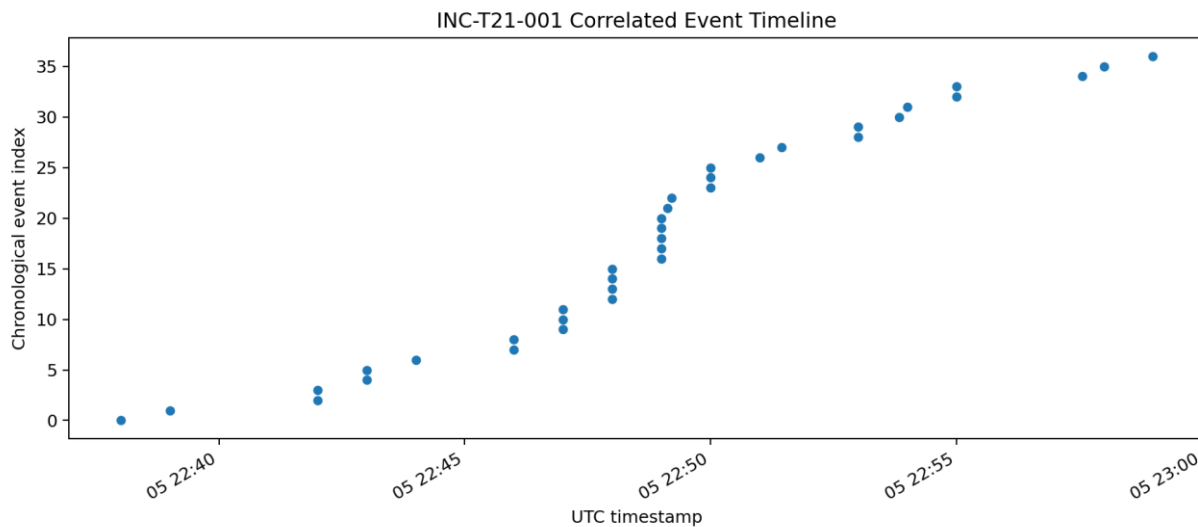


Figure 10. INC-T21-001 correlated event timeline

Event Id	Timestamp Utc	Evidence Source	Observed Action	Analytical Interpretation
VPN-00877	2026-06-05T22:39:00Z	VPN/IAM	SUCCESS vendor session; MFA=FAILED; zone=OT-ZONE; new_device=YES	This session doesn't fit the vendor's normal pattern — it's tied to off-hours timing, an unrecognised device, and a failed or missing MFA check.
EDR-00572	2026-06-05T22:42:00Z	Endpoint	SUSPICIOUS_PROCESS; process=remote_admin_tool; severity=CRITICAL	Endpoint telemetry provides corroborating evidence of unusual remote/administrative tooling on the affected device.



NET-01570	2026-06-05T22:42:00Z	Network	UNUSUAL_OT_CONNECTION; device=DEV0099; bytes_out=6371623.0; dst_port=22.0	Network evidence shows unusual OT/remote-administration connections and/or large outbound transfers from the affected device.
OT-INC-001-01	2026-06-05T22:46:00Z	OT/SCADA	STATUS_READ; asset=CRANE03; state=RUNNING; sequence_valid=YES	CRANE03 reports a running state, but the command sequence around it doesn't check out as valid.
OT-INC-001-02	2026-06-05T22:47:00Z	OT/SCADA	PLC_MODE_CHANGE; asset=CRANE03; state=ABNORMAL; sequence_valid=NO	The PLC mode switch to an abnormal state, combined with an invalid sequence flag, is the first clear sign that something on CRANE03 has gone off-script.
OT-INC-001-03	2026-06-05T22:48:00Z	OT/SCADA	UNAUTHORIZED_COMMAND; asset=CRANE03; state=DEGRADED; sequence_valid=NO	An unauthorized command reaching CRANE03 while the system is already in a degraded state is hard to explain as routine maintenance.
PHY-INC-001	2026-06-05T22:49:00Z	Physical access	GRANTED physical access; badge=BADGE-0014; zone=OT-ZONE; direction=IN	The badge records place a vendor physically inside the restricted OT zone during the same window — someone was on-site, not just connecting remotely.
OT-INC-001-04	2026-06-05T22:49:00Z	OT/SCADA	EMERGENCY_STOP; asset=CRANE03; state=SAFE_STOP; sequence_valid=NO	The emergency stop that follows looks like the crane's own safety logic reacting to the preceding



				unauthorized commands, rather than a planned shutdown.
CARGO-INC-001	2026-06-05T22:51:00Z	Cargo	MANIFEST_CHANGE; container=CONT-004821; manifest_change=YES; destination=DST-Z	A cargo-manifest change lands right in the middle of the same incident window, which is what pulls this from a pure IT/OT story into one with a direct cargo-integrity angle.
EDR-01173	2026-06-05T22:53:00Z	Endpoint	REMOTE_EXECUTION; process=credential_helper_sim; severity=CRITICAL	Endpoint telemetry provides corroborating evidence of unusual remote/administrative tooling on the affected device.
NET-00643	2026-06-05T22:55:00Z	Network	LARGE_OUTBOUND_TRANSFER; device=DEV0099; bytes_out=4134890.0; dst_port=8443.0	Network evidence shows unusual OT/remote-administration connections and/or large outbound transfers from the affected device.

7.3 Evidence assessment

Source	Entity	Observed Evidence	Interpretation
VPN/IAM	V014 / ACC0014 / DEV0099	Off-hours sessions, new device, failed/absent MFA indicators and OT-zone access	Supports unauthorized use of a vendor account.
Endpoint	DEV0099	Remote administration / PLC / credential-helper simulated tools and critical alerts	Corroborates unusual activity on the same device.



OT/SCADA	CRANE03	PLC mode change, unauthorized command, emergency stop; invalid sequence	Shows operational/OT impact during the correlated window.
Network	DEV0099	Unusual OT connections, remote administration activity and large outbound transfer alerts	Corroborates network-level abnormality.
Physical access	BADGE-0014	Vendor badge granted entry to OT-ZONE at 22:49 UTC	Provides physical-context evidence during the incident window.
Cargo	CONT-004821	Manifest destination change at 22:51 UTC	Shows business-process integrity impact.
Maintenance note	CRANE03 / V014	Cross-source proximity noted; incident review initiated	Documents analyst escalation and corroboration.

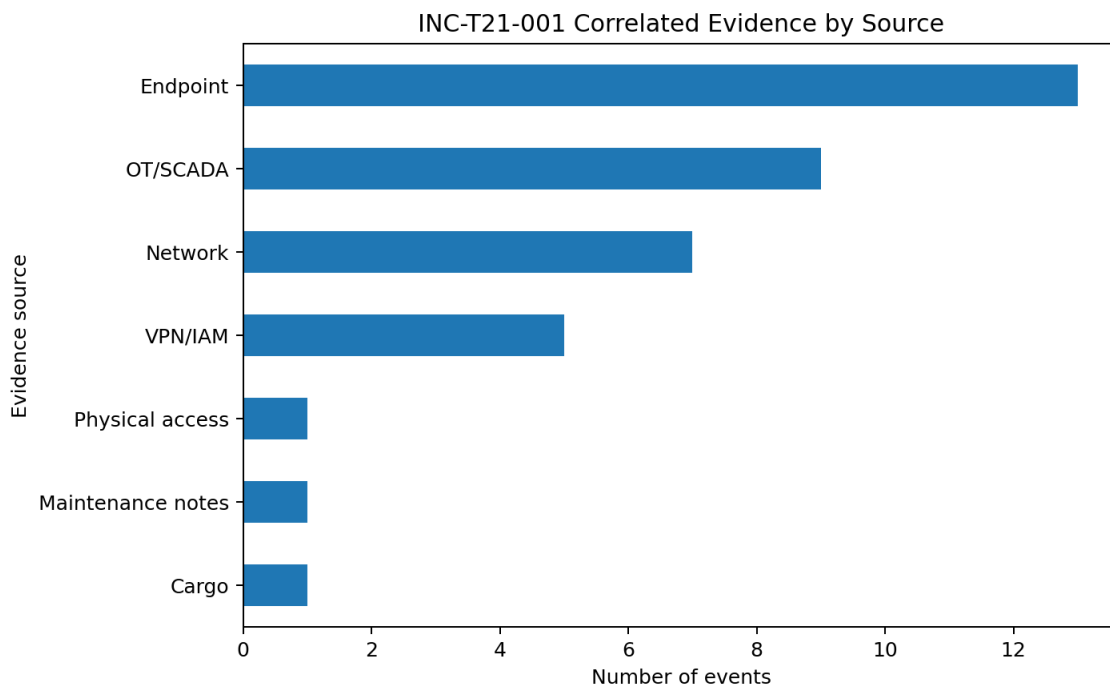


Figure 11. INC-T21-001 evidence volume by source

7.4 Hypothesis assessment

We weighed three competing explanations against the evidence gathered. The confidence labels attached to each one describe how well the synthetic evidence supports it they are not statistical probabilities.



Id	Hypothesis	Evidence Assessment	Confidence
H1	Legitimate maintenance activity	Partially supported by the presence of a vendor session, but weakened by new-device/off-hours/failed-MFA indicators and invalid OT commands.	Low
H2	Equipment fault or operator error	Could explain individual OT alarms, but does not explain the correlated identity, endpoint, network, physical-access and cargo evidence as well.	Low–Moderate
H3	Cyber-enabled vendor-account compromise or unauthorized use	Supported by correlated anomalous vendor access, endpoint remote tooling, invalid OT commands, network anomalies, restricted-zone access and cargo change.	High

Taken together, the cross-source evidence points most strongly toward unauthorized use or compromise of the vendor account. That conclusion rests on the combination of anomalous vendor access, unusual endpoint activity, an invalid OT command sequence, abnormal network traffic, restricted zone badge access and a cargo-manifest change all inside the same short window. What the evidence can't do is identify who was actually behind it.

7.5 Incident response actions

For the response plan, we followed the detection/analysis → containment → eradication/recovery → post-incident review structure from NIST SP 800-61 Rev. 3, applied to this synthetic scenario.

Phase	Response Action	Purpose
Containment	Temporarily suspend the affected vendor account ACC0014 and isolate DEV0099 from OT-connected services.	Limit further unauthorized access while preserving evidence.
Containment	Restrict remote vendor access to the affected OT zone and require verified MFA before reactivation.	Reduce recurrence risk while maintaining controlled vendor support.



Eradication	Review and remove unauthorized remote/administrative tooling represented in endpoint alerts; rotate affected access credentials in the simulation.	Remove the simulated persistence/access path.
Recovery	Validate CRANE03 operating state and review the affected cargo manifest record before returning affected functions to normal operation.	Restore safe and trusted operational state.
Monitoring	Increase monitoring of V014/ACC0014/DEV0099 and correlated OT/network events after recovery.	Detect recurrence or residual abnormal behaviour.
Lessons learned	Tune cross-source correlation rules to combine identity, endpoint, OT, network, physical and cargo signals.	Improve earlier detection of multi-stage events.

7.6 Reproducibility and outputs

The investigation is reproducible from 03_notebooks_or_scripts/08_incident_investigation.py. Outputs are stored in 08_outputs/incident_investigation/. The principal evidence file is incident_correlated_timeline.csv; supporting outputs include incident_summary.csv, primary_incident_evidence_summary.csv, hypothesis_assessment.csv, incident_response_actions.csv and the two investigation figures.

8. Security intelligence

The intelligence module takes what the investigation found and turns it into something usable: priority intelligence requirements, operational indicators, ICS-technique context, and a decision-ready executive summary.

8.1 Priority Intelligence Requirements

pir_id	priority_intelligence_requirement	current_answer	confidence	evidence_basis
PIR-01	Which vendor accounts and devices require elevated investigation priority?	V014 / ACC0014 / DEV0099	High	Supported by correlated VPN/IAM, endpoint, network and incident evidence.



PIR-02	Which OT assets show activity correlated with anomalous remote access?	CRANE03	High	CRANE03 appears in the primary incident evidence and invalid command sequence.
PIR-03	Do remote-access anomalies correlate with operational or business-process changes?	Yes: OT command activity and cargo-manifest change	High	Cross-source timeline shows close temporal correlation in INC-T21-001.
PIR-04	Which indicators should be monitored after containment?	V014, ACC0014, DEV0099, CRANE03 and related OT/network events	High	These entities recur across the primary incident evidence.

8.2 Collection and enrichment

We focused collection on the entities and signals that mattered to the investigation vendor/account/device identity, remote-access details, endpoint alerts, OT command and state changes, network anomalies, physical-access context and cargo-manifest changes. Where the synthetic behaviour lined up with a known ATT&CK for ICS technique, we noted the mapping, but purely as context, not as proof that a real adversary used that technique.

ATT&CK for ICS technique	T21 evidence basis	Evidence reference	limitation
External Remote Services	Remote vendor access is represented in the synthetic VPN/IAM evidence.	INC-T21-001 / VPN-IAM	Contextual match; not proof of adversary technique use.
Valid Accounts	The incident scenario concerns anomalous use of a vendor account.	INC-T21-001 / VPN-IAM	Contextual match; the dataset does not establish how credentials were obtained.
Remote Services	Endpoint/network evidence includes simulated remote-administration activity.	INC-T21-001 / Endpoint + Network	Contextual match to the observed behaviour.



Change Program State	OT evidence records PLC mode/state changes during the incident window.	INC-T21-001 / OT-SCADA	Relevant ICS technique context; synthetic event does not establish a real-world adversary.
Modify Parameter	OT evidence contains abnormal command/process-state changes.	INC-T21-001 / OT-SCADA	Contextual enrichment only; command semantics are synthetic.

ATT&CK for ICS gives us an ICS-specific vocabulary for these techniques like External Remote Services, Valid Accounts, Remote Services, Change Program State and Modify Parameter all showed up as relevant reference points for what we observed.

8.3 Operational intelligence product

indicator_type	indicator	context	source incident	confidence
ENTITY	V014	Vendor identity associated with the primary incident	INC-T21-001	High
ACCOUNT	ACC0014	Vendor account associated with anomalous access	INC-T21-001	High
DEVICE	DEV0099	Device associated with endpoint and network anomalies	INC-T21-001	High
OT_ASSET	CRANE03	OT asset associated with invalid command sequence	INC-T21-001	High
BADGE	BADGE-0014	Physical-access identifier observed in restricted OT zone	INC-T21-001	Medium
CARGO	CONT-004821	Cargo record with correlated manifest change	INC-T21-001	Medium
operational_priority	entity_or_control_area	analyst_action	priority	
Immediate investigation	V014 / ACC0014 / DEV0099	Review remote-access history, endpoint alerts and related OT/network events.	High	
OT monitoring	CRANE03	Increase monitoring of command sequences, mode/state changes and	High	



		associated remote sessions.	
Access control	Vendor remote access	Require verified MFA and controlled access before restoration of affected vendor connectivity.	High
Business integrity	CONT-004821	Validate the affected manifest record and any downstream cargo-processing changes.	Medium

8.4 Executive intelligence product

executive_topic	finding	confidence
Risk picture	This investigation shows that identity, endpoint, OT, network, physical and cargo signals genuinely can be pulled together into one coherent incident picture, rather than sitting as separate, harder-to-interpret alerts.	High
Primary concern	INC-T21-001 is the clearest example: anomalous vendor access lining up with OT and business-process indicators, all within a short window.	High
Control implication	For this environment, the controls that matter most are tighter vendor remote-access management, stronger MFA, cross-source monitoring, and correlation that's specifically aware of OT context.	High
Decision support	Our recommendation to an analyst: prioritise investigation and containment whenever remote-access anomalies show up at the same time as OT state/command changes or anything touching cargo integrity.	High

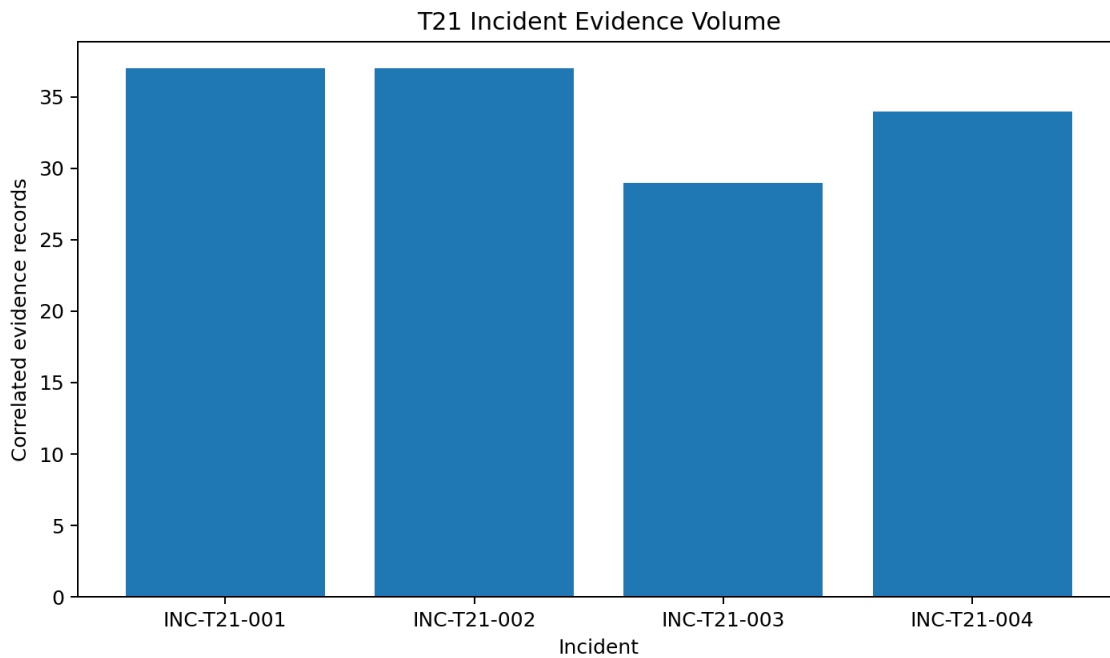


Figure 12. T21 incident evidence volume used for intelligence prioritisation

8.5 Intelligence feedback loop

feedback_area	lesson_from_analysis	collection_or_detection_change
Detection	Cross-source correlation identified a stronger signal than any single source.	Retain correlated identity + endpoint + OT + network + physical + cargo fields in collection.
Access	Vendor access anomalies were central to the primary investigation.	Prioritise MFA status, device novelty, access zone and off-hours features.
OT	Invalid command sequences provided operational context.	Collect command sequence validity and process-state transitions for future analytics.
Business integrity	Cargo-manifest change added business impact context.	Continue collecting manifest-change events for incident correlation.

Feeding this back into collection priorities, we'd want future detection to keep an eye on vendor-access context, MFA status, device novelty, OT command-sequence validity, process-state transitions and cargo-manifest changes.



8.6 Implementation evidence

The security intelligence implementation is reproducible from 03 notebook or scripts/10_security_intelligence.py. Outputs are stored in 08_outputs/security_intelligence/. The implementation produced four PIRs, six operational indicators, five contextual ATT&CK for ICS mappings, an operational intelligence product, an executive intelligence product and a documented feedback loop. The QA output is security_intelligence_qa.csv.

9. Simulation and security-control evaluation

For the simulation, we built a Monte Carlo what-if model to compare alternative security controls against the four synthetic incident scenarios. It estimates how each control shifts expected incident frequency, our analytical risk index, and continuity impact over a 30-day window. It's worth being clear that the effectiveness numbers we used are modelling assumptions we set, not measured real-world control performance.

This simulation is decision support for the synthetic T21 environment, it isn't a forecast of what would actually happen at a real port.

9.1 Simulation model and assumptions

Assumption	Value
Baseline incident frequency	One observed synthetic incident per scenario in the 30-day dataset; Poisson lambda=1.0
Risk weights	S01=8, S02=10, S03=9, S04=7; analytical scale only
MFA/session restriction effectiveness	65% occurrence reduction; 25% impact reduction before scenario modifiers
Network segmentation effectiveness	45% occurrence reduction; 50% impact reduction before scenario modifiers
Enhanced monitoring effectiveness	20% occurrence reduction; 55% impact reduction before scenario modifiers
Continuity multipliers	MFA=1.10, segmentation=1.25, monitoring=1.05
Simulation iterations	10,000 Monte Carlo iterations with seed 210821

The baseline incident frequency comes straight from how we built the dataset: each of S01–S04 has exactly one injected incident in the 30-day period, so we modelled that as a Poisson process with $\lambda=1$ per scenario and resampled repeatedly. The impact weights (S01=8, S02=10, S03=9, S04=7) are just an analytical risk scale we defined, and the continuity-hour figures represent the operational disruption we assumed for each scenario. We documented all of this so the simulation can be rerun later with real measured parameters if they ever become available.

9.2 Control scenarios

Scenario	Control implementation	Primary effect
Baseline	No additional control; current synthetic incident rates and impact assumptions.	Reference case.



MFA + session restrictions	Require stronger vendor authentication and restrict unusual/unsupported sessions.	Targets account misuse and remote-access abuse.
Network segmentation	Restrict communication paths between vendor access, IT/network and OT zones.	Targets unauthorized reach into OT assets and limits blast radius.
Enhanced monitoring	Increase cross-source detection and early analyst response for high-risk vendor/OT activity.	Primarily reduces impact through earlier detection and response.

9.3 Comparative simulation results

Control	Mean events / 30 days	95th percentile events	Mean risk index	95th percentile risk index	Mean continuity hours	95th percentile continuity hours	Risk reduction vs baseline (%)
Baseline	4.04	8.00	34.41	65.00	11.64	22.00	0.00
MFA + session restrictions	1.67	4.00	11.19	27.62	5.45	13.20	67.49
Network segmentation	2.51	5.00	14.36	31.20	8.77	19.38	58.27
Enhanced monitoring	3.20	6.00	19.25	38.76	9.68	19.95	44.06

Under these assumptions, baseline comes out at a mean risk index of 34.41. Adding MFA and session restrictions brings that down to 11.19 — a 67.50% reduction while network segmentation gets to 14.36 (58.27% reduction) and enhanced monitoring to 19.25 (44.06% reduction). All three controls also lower expected continuity impact: 5.45 hours for MFA/session restrictions, 8.77 for segmentation and 9.68 for monitoring, against 11.65 hours at baseline.

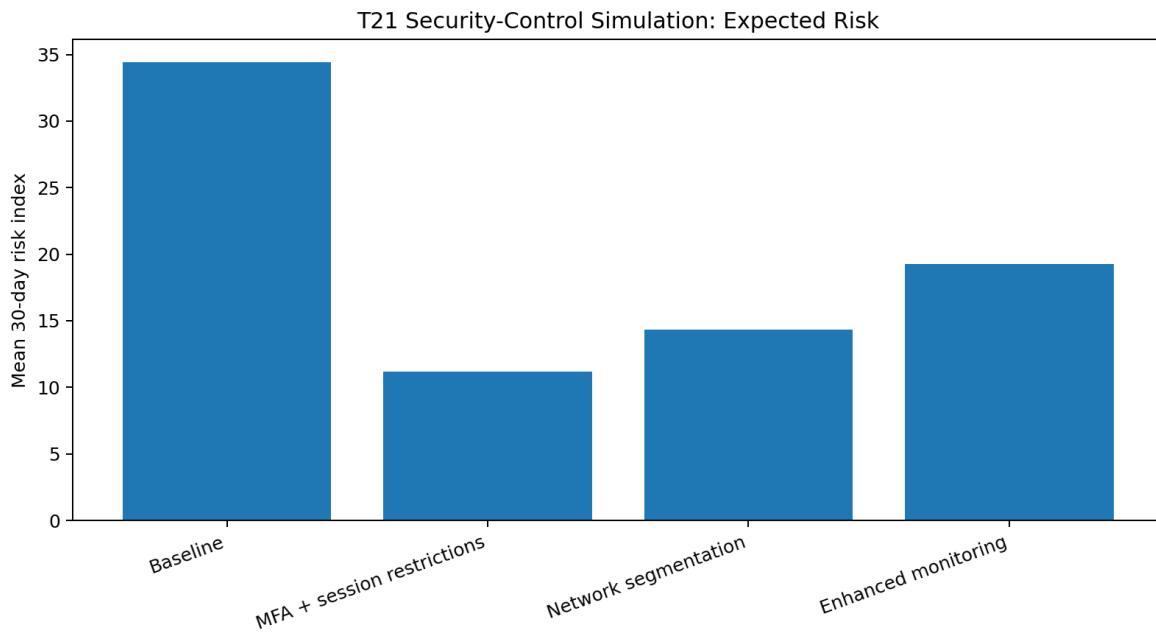


Figure 13. Simulated mean 30-day risk index by security-control scenario

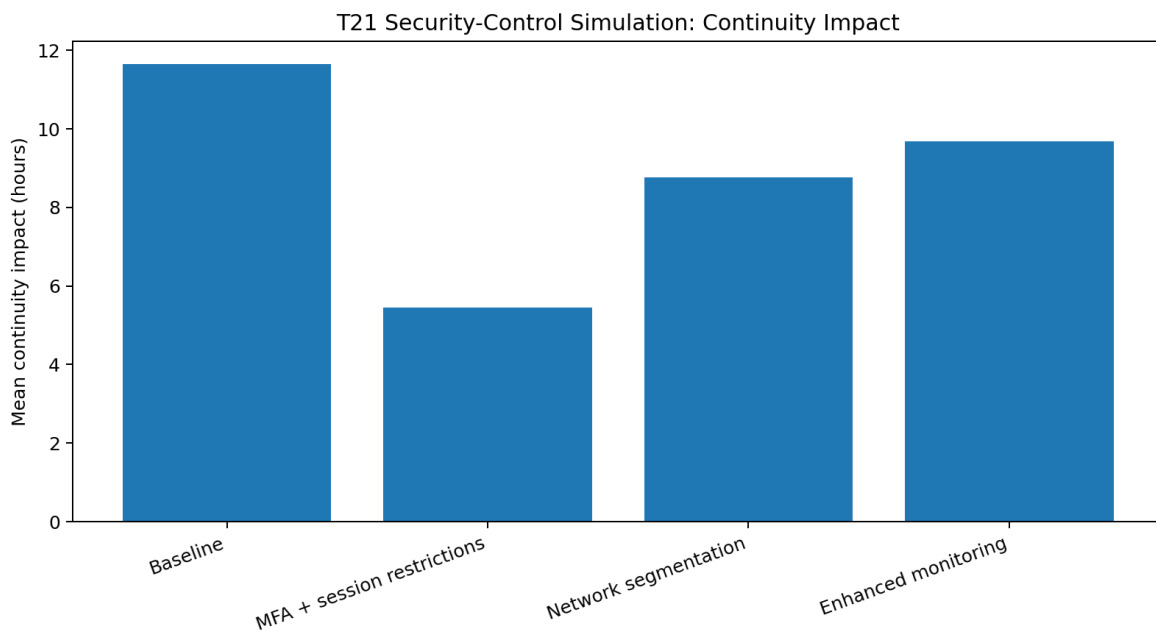


Figure 14. Simulated mean continuity impact by security-control scenario



9.4 Scenario-level comparison

Control	Scenario ID	Scenario	Expected events / 30 days	Impact factor	Expected risk index
Baseline	S01	Compromised vendor account	1.000	1.000	8.000
Baseline	S02	Unauthorized OT command sequence	1.000	1.000	10.000
Baseline	S03	Remote administration and data transfer	1.000	1.000	9.000
Baseline	S04	Cargo/manifest manipulation	1.000	1.000	7.000
MFA + session restrictions	S01	Compromised vendor account	0.350	0.775	2.170
MFA + session restrictions	S02	Unauthorized OT command sequence	0.512	0.800	4.100
MFA + session restrictions	S03	Remote administration and data transfer	0.350	0.762	2.402
MFA + session restrictions	S04	Cargo/manifest manipulation	0.448	0.775	2.428
Network segmentation	S01	Compromised vendor account	0.752	0.600	3.612
Network segmentation	S02	Unauthorized OT command sequence	0.550	0.775	4.263
Network segmentation	S03	Remote administration and data transfer	0.550	0.725	3.589
Network segmentation	S04	Cargo/manifest manipulation	0.640	0.625	2.800
Enhanced monitoring	S01	Compromised vendor account	0.800	0.698	4.464
Enhanced monitoring	S02	Unauthorized OT command sequence	0.800	0.698	5.580
Enhanced monitoring	S03	Remote administration and data transfer	0.800	0.752	5.418

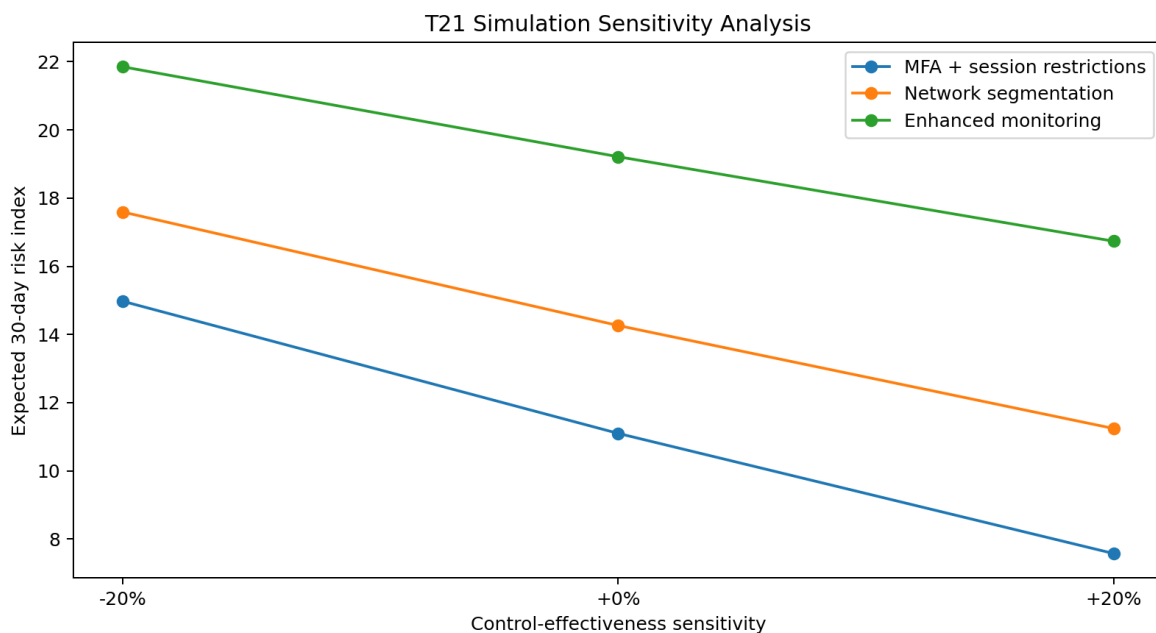


Enhanced monitoring	S04	Cargo/manifest manipulation	0.800	0.670	3.752
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Breaking it down by scenario, MFA and session restrictions do the most to blunt the compromised-vendor-account scenario specifically, while segmentation is what really cuts into the unauthorized-OT-command scenario. Enhanced monitoring helps everywhere by shortening how long an issue goes unnoticed, but it doesn't close off the access path itself. Again, these are what the simulation shows under our assumptions, not measured control effectiveness.

9.5 Sensitivity analysis

Control	Effectiveness adjustment	Expected risk index	Risk reduction vs baseline (%)
MFA + session restrictions	-20%	14.97	56.48
MFA + session restrictions	+0%	11.10	67.74
MFA + session restrictions	+20%	7.58	77.98
Network segmentation	-20%	17.59	48.87
Network segmentation	+0%	14.26	58.55
Network segmentation	+20%	11.24	67.33
Enhanced monitoring	-20%	21.85	36.50
Enhanced monitoring	+0%	19.21	44.16
Enhanced monitoring	+20%	16.74	51.36



**Figure 15. Sensitivity of expected risk to assumed control effectiveness**

We stress tested the assumptions by shifting control effectiveness $\pm 20\%$. The direction never flips: weaker assumed effectiveness always pushes expected residual risk up, stronger always pulls it down. That's reassuring in one sense, but it also means our comparative results are only as good as the effectiveness numbers we assumed, and should be revisited if real data ever becomes available.

9.6 Implementation evidence and limitations

0	1	2
Simulation model	Monte Carlo what-if simulation with 10,000 iterations and fixed seed 210821.	12_simulation.py
Control scenarios	Baseline plus three alternative controls: MFA/session restrictions, network segmentation and enhanced monitoring.	scenario_control_comparison.csv
Comparative results	Expected events, risk index and continuity impact for each control scenario.	control_simulation_summary.csv
Sensitivity	Control-effectiveness variation of -20%, 0% and +20%.	simulation_sensitivity.csv
Visual evidence	Three simulation figures generated from the same results tables.	08_outputs/simulation/

To be clear about the limits here: this simulation doesn't model live network traffic, actual crane control logic, or real operational consequences. The control effectiveness figures, risk weights and continuity numbers are all synthetic assumptions, not empirical measurements. What the module does give us is a transparent, reproducible way to compare options for the T21 decision problem.

10. Text mining/NLP

For the text-mining side, we worked with the 304 synthetic maintenance and incident notes generated for T21. The corpus draws from 11 unique templates and breaks down into 236 NORMAL, 64 SUSPICIOUS and 4 INCIDENT records. We collapsed this into a binary target for the classifier SUSPICIOUS and INCIDENT count as risk (1), NORMAL as non-risk (0).

Table 10.0 — Text-mining/NLP implementation summary

Required element	Implementation evidence
Text corpus	304 synthetic maintenance-note records; 236 NORMAL, 64 SUSPICIOUS and 4 INCIDENT; 11 unique text templates.
Preprocessing	UTC timestamps parsed; free-text nulls handled; TF-IDF unigrams/bigrams with English stop-word removal and sublinear TF.



Required element	Implementation evidence
Entities/indicators/topics/classes	Top risk terms include remote, unexpected, remote session, outside, vendor remote and outside approved; entity extraction identifies vendor, crane, PLC, yard and note identifiers.
Evaluation/validation	Stratified 70/30 split with seed 210821. Accuracy=1.0000; precision=1.0000; recall=1.0000; F1=1.0000; false-negative rate=0.0000; TN=71, FP=0, FN=0, TP=21.
Operational use	Scores all 304 notes and exports a ranked top-15 list for analyst triage; text indicators are decision-support evidence, not standalone incident declarations.
Limitations	The corpus has only 11 unique text templates and is synthetic. Perfect metrics are therefore dataset-specific and are not generalized to real-world prose.

10.1 Corpus and preprocessing

We loaded the corpus from `t21_maintenance_notes.csv`, converted timestamps to UTC, and filled any missing free text with empty strings. For features we used TF-IDF over unigrams and bigrams, with English stop words removed and sublinear term frequency scaling. The same 70/30 stratified split (seed 210821) gave us 212 training notes and 92 test notes.

10.2 Text classification model and evaluation

The classifier itself is TF-IDF features feeding a logistic regression with balanced class weights, evaluated only on the held-out test set. The confusion matrix came out TN=71, FP=0, FN=0, TP=21 — accuracy, precision, recall and F1 all landed at 1.00, with zero false negatives. We're clear-eyed about why: this corpus only has 11 unique templates, so a perfect score here says more about how repetitive the synthetic text is than it does about how the model would handle messy real-world notes.

Metric	Result
Training records	212
Test records	92
Accuracy	1.0000
Precision	1.0000
Recall	1.0000
F1-score	1.0000
False-negative rate	0.0000
True negatives	71
False positives	0
False negatives	0



True positives	21
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10.3 Text indicators and operational use

The terms the logistic regression weighted most heavily toward risk were things like 'remote', 'unexpected', 'remote session', 'outside', 'vendor remote', 'observed outside' and 'outside approved' which line up sensibly with the kind of language you'd expect around unauthorized remote access. These give an analyst transparent, explainable indicators rather than a black-box verdict. We scored all 304 notes and ranked the highest-risk ones for review.

Term	Coefficient
remote	1.3601
unexpected	0.8889
remote session	0.8889
outside	0.8889
vendor remote	0.8889
unexpected vendor	0.8889
observed outside	0.8889
outside approved	0.8889
session observed	0.8889
vendor	0.8060

10.4 Text-risk outputs

At the 0.50 threshold, 68 of the 304 notes came back flagged. We exported both the full scored corpus and a top-15 ranked list and also pulled out entity references from the text itself vendor, crane, PLC and yard identifiers, plus note IDs, wherever they appeared.

Figure 16 — Text corpus risk-class distribution

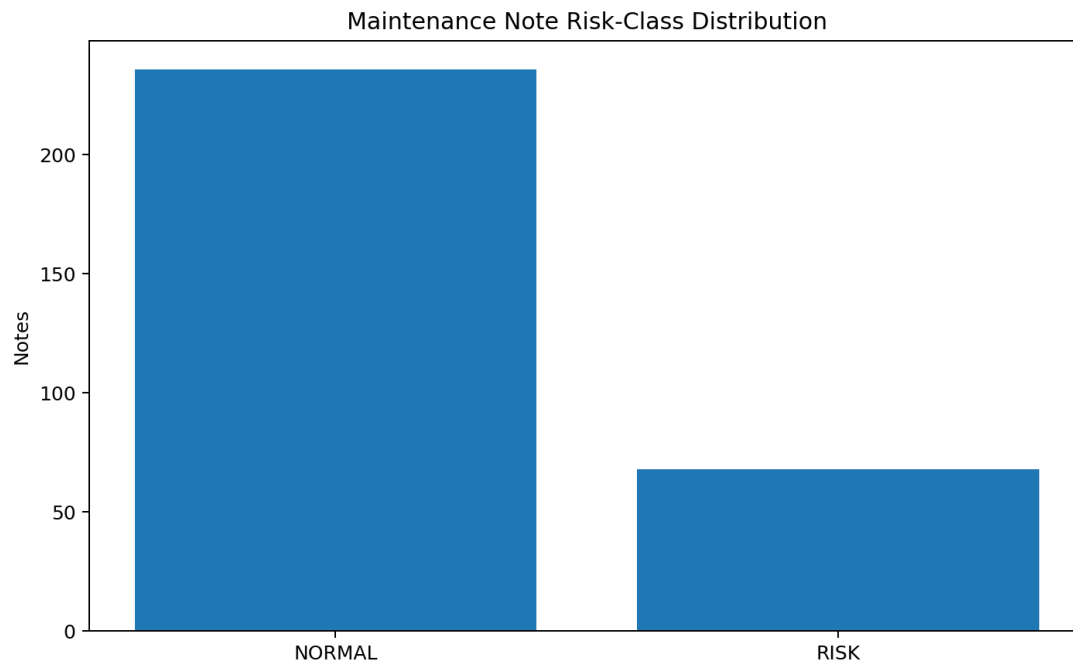


Figure 17 — Top TF-IDF terms associated with risk

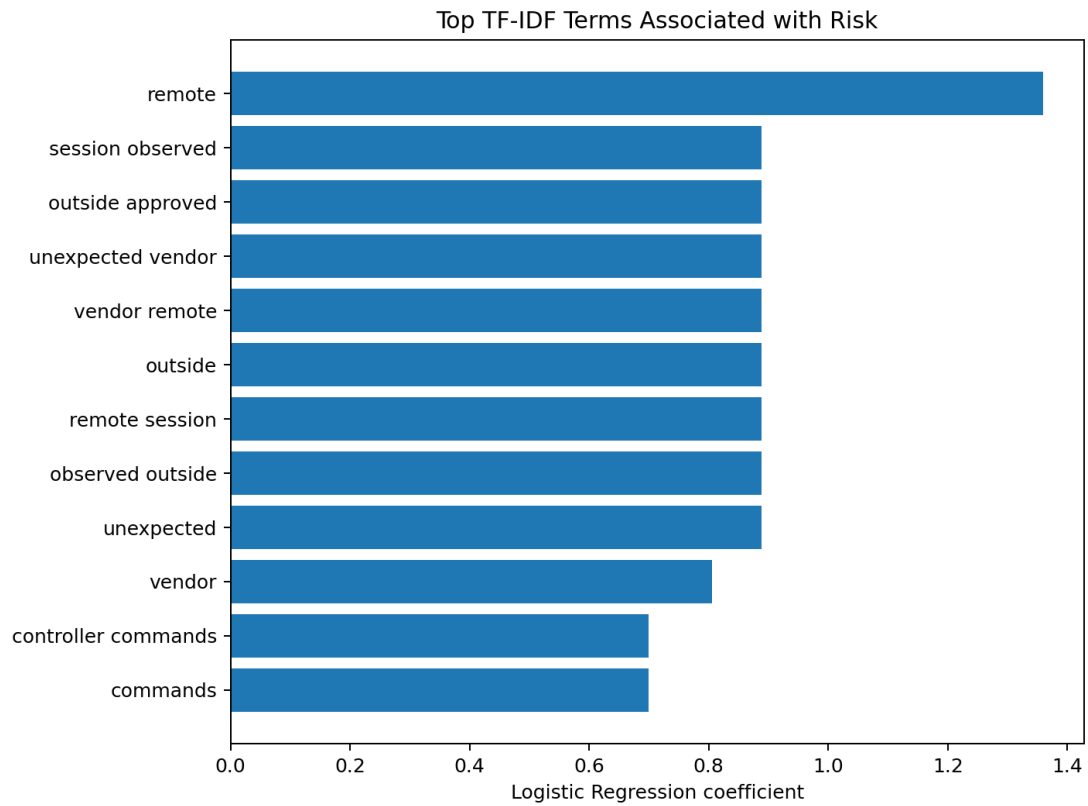
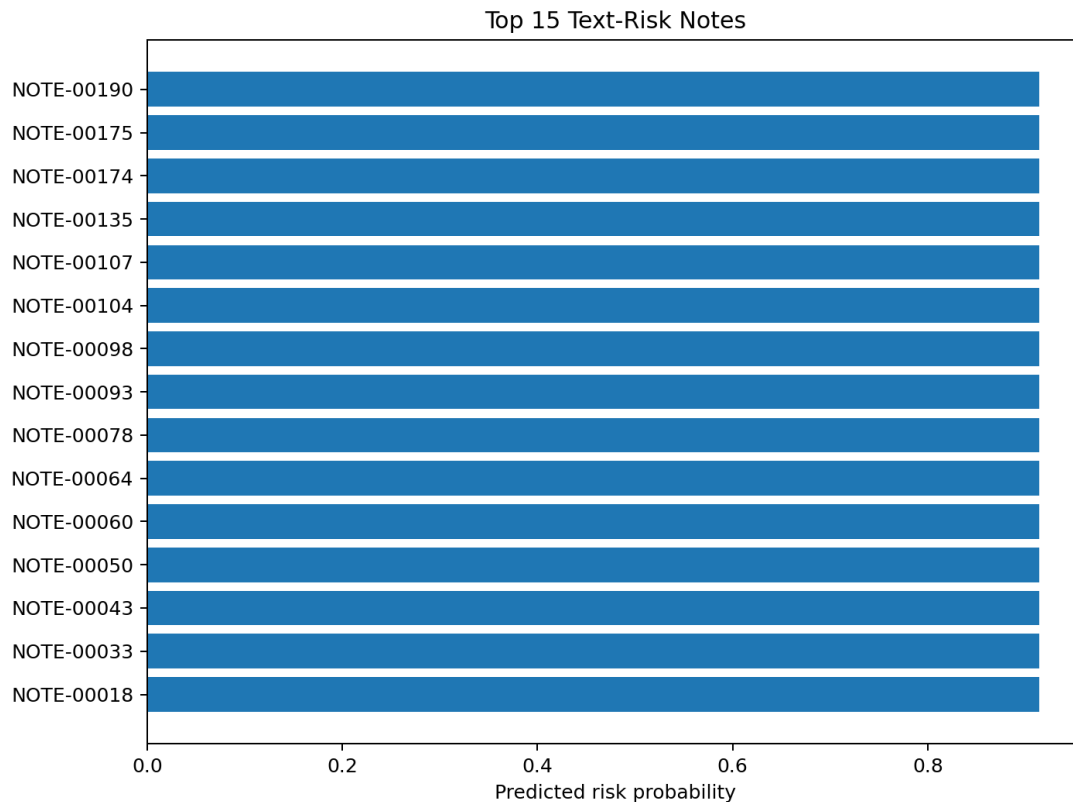


Figure 18 — Top 15 text-risk notes



10.5 Operational interpretation and limitations

In practice, this module helps an analyst triage maintenance narrative by turning free text into a repeatable risk score with transparent term level reasoning behind it flagging notes that mention unexpected vendor activity, remote sessions, or access outside approved windows. But we don't want to oversell it: with only 11 unique templates, this corpus is highly repetitive, and the perfect held-out scores reflect that repetition rather than genuine language understanding. This stays decision support evidence for an analyst, not something that declares an incident on its own.

Implementation evidence: 14_text_mining_nlp.py; nlp_model_metrics.csv; nlp_confusion_matrix.csv; top_15_text_risk_notes.csv; top_risk_terms.csv; text_entities.csv; and the three NLP figures in 08_outputs/text_mining/.

11. Predictive and adversarial analysis

The predictive module looks at the risk of a vendor session using only information that would be available before or right at the start of that session, nothing from the future. We used a chronological 70/30 split, so the test period genuinely comes after the training period, and built prior-24-hour account history features without letting any current or future label leak in. The result is a forward-looking risk signal, a deliberately different setup from the random split used in the earlier supervised detector.



11.1 Predictive model design

Target: binary non-normal session risk (SUSPICIOUS/INCIDENT-equivalent = 1, NORMAL = 0). What we're forecasting: the risk of the current session, based only on prior account behaviour and session-start context. Features cover vendor/account identity, role, access zone, authentication result, MFA status, off-hours status, new-device status, plus prior-24-hour history: session count, non-normal count/rate, authentication failures, off-hours sessions, new-device sessions, mean session duration, and time since the last session.

Same Random Forest setup as before 300 trees, max depth 10, minimum leaf 2, balanced class weights, seed 210821, threshold at 0.50. The chronological split left us with 1,260 training sessions and 540 test sessions.

Table 11.1 — Predictive model evaluation results

Measure	Result
Training sessions	1260
Test sessions	540
Accuracy	95.56%
Precision	53.66%
Recall	81.48%
F1-score	64.71%
False-negative rate	18.52%
Threshold	0.50
False positives	19
False negatives	5

Test performance came out to 95.56% accuracy, 53.66% precision, 81.48% recall and an F1-score of 64.71%, with an 18.52% false-negative rate. The lower precision tells us the model raises a fair number of false alarms, which is exactly why this is built for analyst triage, not automatic containment.

11.2 Predictive risk ranking

Table 11.2 — Top 15 predictive-risk sessions

event id	vendor id	account id	risk probability	target
VPN-01652	V008	ACC0050	89.87%	NON-NORMAL
VPN-00419	V014	ACC0052	89.33%	NON-NORMAL



VPN-01421	V003	ACC0035	88.87%	NON-NORMAL
VPN-00780	V019	ACC0027	88.28%	NON-NORMAL
VPN-00399	V019	ACC0047	83.57%	NON-NORMAL
VPN-00441	V020	ACC0034	83.53%	NON-NORMAL
VPN-00590	V010	ACC0004	80.2%	NORMAL
VPN-01164	V015	ACC0059	77.78%	NON-NORMAL
VPN-01544	V011	ACC0044	75.73%	NON-NORMAL

This ranking gives an analyst a queue of sessions worth checking first. A high predicted probability just means the session looks similar to non-normal patterns we've seen before in the synthetic data it isn't a verdict that any particular vendor or user is acting maliciously.

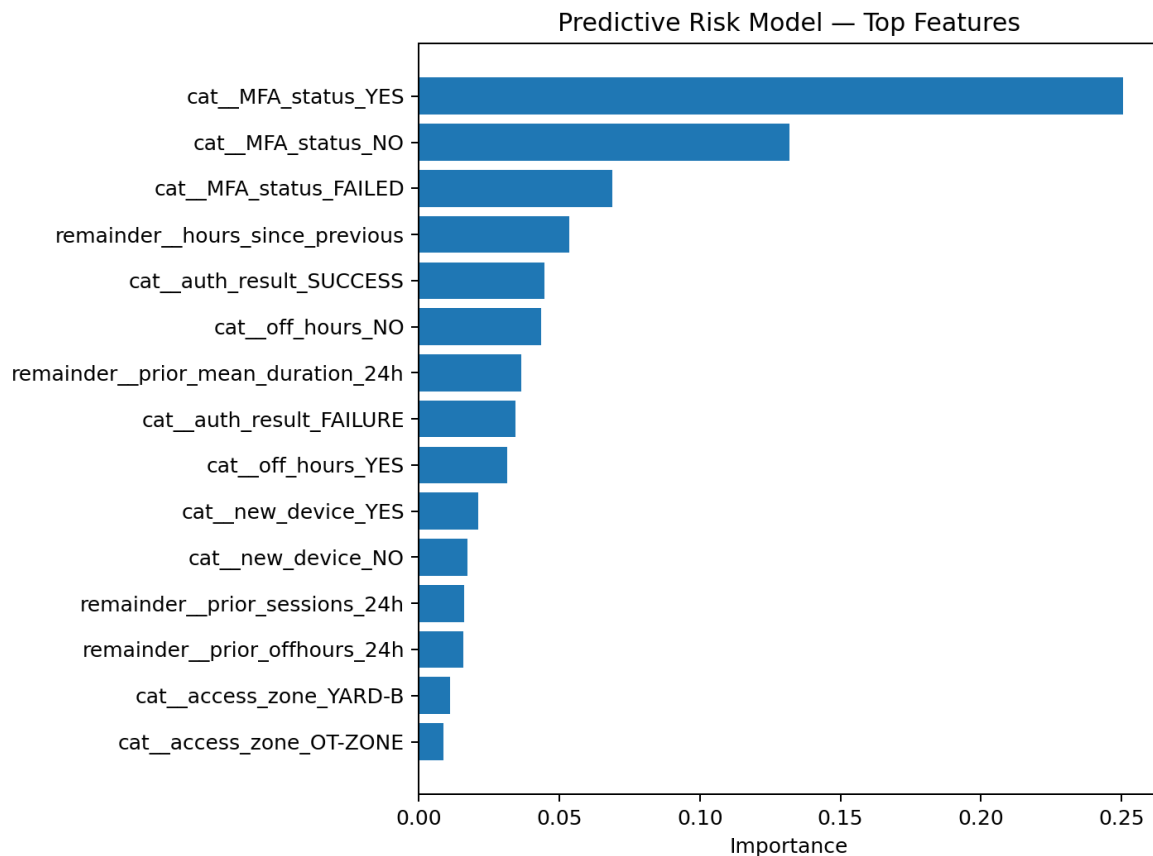


Figure 19 — Top features used by the predictive risk model

11.3 Adversarial stress testing

We ran three controlled adversarial tests against the three highest-risk held-out sessions, purely to probe model sensitivity none of this touched a live system. The three cases were: masking temporal/device/MFA indicators, swapping in benign-looking context, and diluting the prior-history signal. This approach follows



NIST's adversarial machine-learning taxonomy, which treats evasion-style manipulation of model inputs as a genuine security concern and pushes for robustness and lifecycle risk management.

Table 11.3 — Adversarial stress-test results

event_id	scenario	original_probability	manipulated_probability	delta	risk_class_manipulated
VPN-01652	A_feature_masking	89.87%	7.88%	- 81.99 %	LOW_RISK
VPN-01652	B_context_substitution	89.87%	85.31%	- 4.56%	HIGH_RISK
VPN-01652	C_history_dilution	89.87%	89.87%	0.0%	HIGH_RISK
VPN-00419	A_feature_masking	89.33%	28.86%	- 60.47 %	LOW_RISK
VPN-00419	B_context_substitution	89.33%	83.35%	- 5.98%	HIGH_RISK
VPN-00419	C_history_dilution	89.33%	90.2%	0.87%	HIGH_RISK
VPN-01421	A_feature_masking	88.87%	3.37%	- 85.49 %	LOW_RISK
VPN-01421	B_context_substitution	88.87%	84.63%	- 4.23%	HIGH_RISK
VPN-01421	C_history_dilution	88.87%	88.87%	0.0%	HIGH_RISK

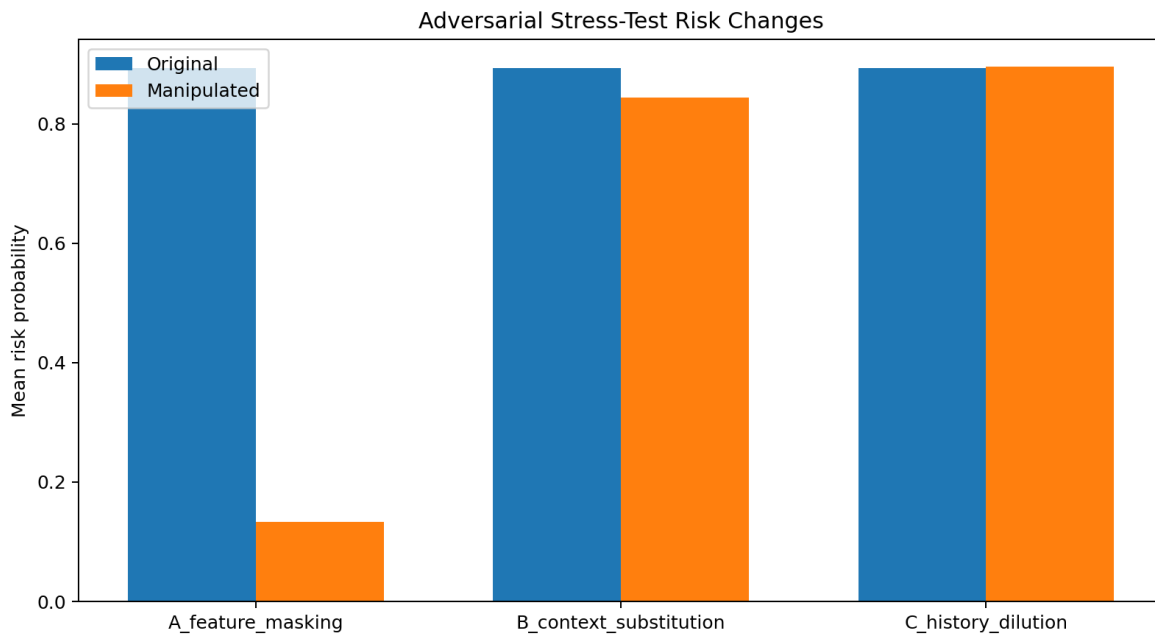


Figure 20 — Change in predictive risk under synthetic adversarial manipulation

Masking features had by far the biggest effect it dropped all three sessions from HIGH_RISK to LOW_RISK. Swapping in benign context barely moved the needle; all three stayed HIGH_RISK. Diluting prior history did almost nothing for these particular cases, which tells us the model leans heavily on current session-level indicators rather than history. The takeaway for us: a single predictive model can be pushed around by the right kind of manipulated input, which is exactly why we're not relying on it alone the anomaly analytics, cross-source correlation and a human in the loop all still matter.

11.4 Governance, drift and limitations

This predictive model is scoped to the synthetic T21 dataset it isn't ready to be called a production risk model. Before any real deployment, monitoring would need to track feature drift, changes in class prevalence, false-negative rates and alert volume, and retraining would need to be version-controlled and checked against a time-separated validation set. Because our adversarial tests use synthetic manipulation, they tell us about model sensitivity, not the actual likelihood of a real attack.

Implementation evidence: 16_predictive_adversarial.py; predictive_model_metrics.csv; predictive_test_scored.csv; top_15_predictive_risk_sessions.csv; predictive_feature_importance.csv; adversarial_stress_test_results.csv; predictive_adversarial_qa.csv; and the two figures in 08_outputs/predictive_adversarial/.

12. Prototype and analyst workflow

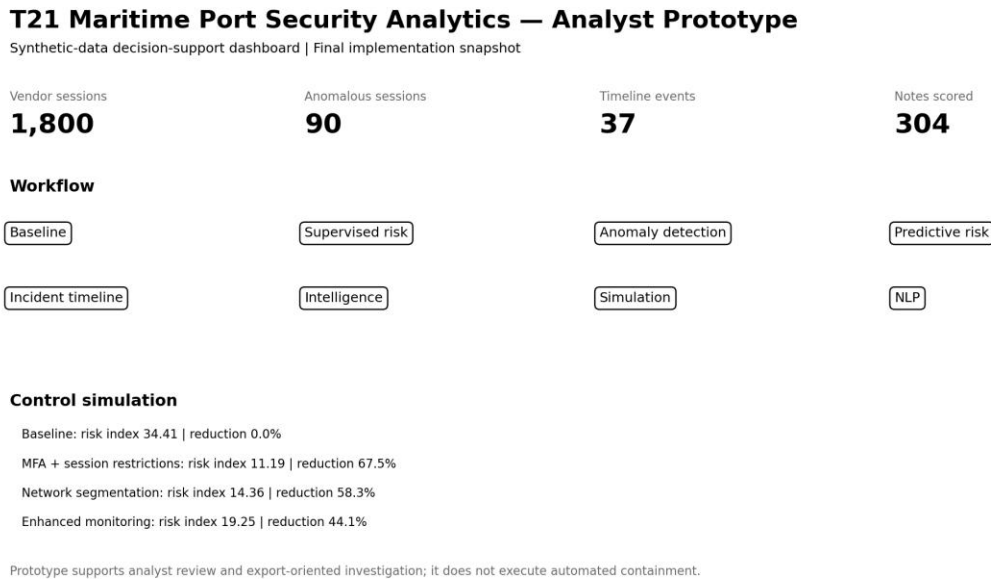
Workflow stage	Implemented function	Primary output
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Load / refresh	Loads generated analytical snapshots and refreshes the browser view	Current analytical snapshot
Baseline and EDA	Displays baseline measures and behavioural context	Baseline summary
Supervised model	Shows top high-risk sessions and predicted labels	Top 15 supervised sessions
Anomaly analytics	Shows anomalous sessions and anomaly scores	Top 15 anomalous sessions
Predictive risk	Displays predictive risk ranking	Top 15 predictive sessions
Incident investigation	Filters and displays correlated evidence timeline	37-event timeline
Security intelligence	Displays operational indicators and confidence	Operational intelligence product
Simulation	Displays control-scenario risk and continuity comparison	Control simulation summary
NLP	Displays high-risk text notes and risk scores	Top 15 text-risk notes

The prototype we built is a self-contained, browser-based dashboard generated from the project's own outputs. It gives an analyst one place to check baseline behaviour, supervised and anomaly risk, predictive risk, correlated incidents, intelligence products, simulation results and NLP findings. It only ever works with synthetic data and doesn't trigger any containment or operational action on its own.

It's generated by `03_notebooks_or_scripts/18_prototype.py`, which writes out `07_dashboard_or_prototype/t21_analyst_prototype.html`. The page embeds the latest analytical snapshot, lets an analyst filter high-risk sessions and incident evidence, and includes a refresh control for when the underlying outputs are regenerated.

**Figure 21: Implemented analyst prototype overview**

To be upfront about its limits: this is a local, static prototype, not a production web app. Data is embedded straight from the generated CSVs, there's no authentication or role-based access built in, and nothing in the interface triggers an automated response. For a capstone built on synthetic data, we think that's the right level of scope.

13. Testing and quality assurance

We tested the finished implementation using functional checks, evidence integrity checks and reproducibility checks all based on the actual files and outputs the project produces, not just what we intended to build.

Final result: 5 of 5 documented acceptance tests passed. A separate internal reproducibility check confirmed all nine major analytical/prototype scripts are present and run. Evidence for both is stored in 08_outputs/final_qa/.

14. Updated schedule and backlog

We're at roughly 90% complete overall.

**15. Team contribution evidence**

Member	Components completed	Repository/artefact evidence	Partner review completed
Member A 215103238 Mwandingi	Data generation and validation; baseline/EDA; supervised and anomaly modelling; predictive/adversarial modelling; simulation; prototype integration and QA support.	03_notebooks_or_scripts/00_generate_t21_dataset.py; 01_validate_dataset.py; 03_baseline_eda.py; 05_supervised_model.py; 07_anomaly_access_analytic s.py; 12_simulation.py; 16_predictive_adversarial.py; 18_prototype.py; 19_final_qa.py.	Completed
Member B 216086728 Kompeli	Incident investigation; security intelligence; NLP/text mining; response-action analysis; intelligence products and prototype content review.	03_notebooks_or_scripts/08_incident_investigation.py; 10_security_intelligence.py; 14_text_mining_nlp.py; incident/intelligence/NLP outputs; prototype review artefacts.	Completed